CHAPTER 1 · GRADE 4 · WEEK 1

Energy Causes Change

Discovery / Evidence · Where do we see energy at work?



Unit 1 · Energy Transfer & Transformation

TURN & TALK BQ

Big Question

Where can we observe evidence of energy causing change?

Keep it in mind all chapter. We will come back to it.

Think: What is ONE change you have seen happen?

WRITE N

BQ

Big Question – My first thinking

Guided Notes p. 1

Where can we observe evidence of energy causing change?

My first thinking

Write 1–2 quick ideas. No wrong answer yet.

What changed?

Reader p. 1 — you walk into a dark room and flip a switch.

DARK room

quiet · hard to see

BRIGHT room

light fills the room

Something became different. That is a change.

I notice… I wonder…

I NOTICE

The room is different now.
Light is everywhere.

I WONDER

What made the light change so fast?

Share one notice and one wonder with a partner.

Electric Pencil Sharpener

Observe the pencil BEFORE we sharpen it.

BEFORE

A dull pencil — tip is flat, and it is quiet.

Look closely. What do you see? What do you hear?

Predict: what will change?

Think before we turn it on.

When the sharpener runs, the pencil will…

get _____

Make a prediction. Say it to your partner.

What changed?

BEFORE

dull tip · quiet

AFTER

shorter · sharper · humming

CHANGES

- the tip is sharper
- the pencil is shorter
- it makes a sound

How do we know energy was involved?

Energy can be invisible. We look for the change.



The pencil got shorter and sharper and made a sound.

Those changes are evidence that energy was involved.

V

Vocabulary — Energy & Change

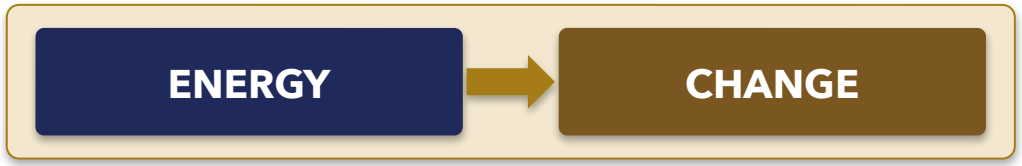
Reader p. 1

VOCABULARY

TEACH

energy

the ability to cause change



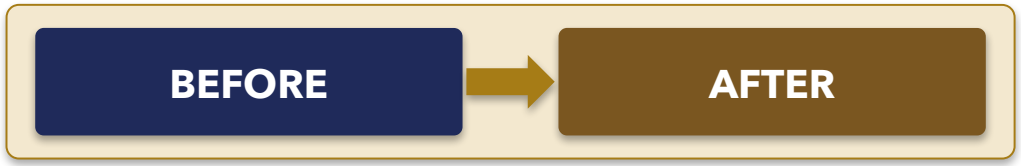
Reader p. 1

VOCABULARY

TEACH

change

to become different



Reader p. 1

WRITE N

V1

Vocabulary – Energy & Change

Guided Notes · Reader p. 1

energy – the ability to cause _____ .

change – to become _____ .

Write the missing word in each blank.

CHECK N

V1

Vocabulary – Energy & Change

Guided Notes · Reader p. 1

energy – the ability to cause **change** .

change – to become **different** .

change · different

RECITE R

What is energy?

Reader pp. 1–2

Energy is the ability to cause change.

Say it together: “Energy is the ability to cause change.”

Change is evidence of energy

Reader p. 2 — the seashore changes all around us.



When you see a change — water moving, sand shifting, sound —

you are seeing evidence of energy.

N1

Energy Causes Change

Guided Notes · Reader pp. 1–2

Energy has the ability to _____ _____ .

When you see a change, you are seeing evidence of energy.



A change is _____ that energy is involved.

N1

Energy Causes Change

Guided Notes · Reader pp. 1–2

Energy has the ability to **cause** **change** .

When you see a change, you are seeing evidence of energy.



A change is **evidence** that energy is involved.

cause · change · evidence

R

Energy has different forms

Reader p. 3 — think of a spinning carousel.

motion | **sound** | **light** | **heat** | **electrical energy**

The carousel spins (motion), plays music (sound), shines bright bulbs
(light), warms those bulbs (heat), and plugs in (electrical energy).

Clap three times

What forms of energy do you observe?

When you clap...

motion — your hands move
sound — you hear a clap
heat — your hands may feel warm

N2

Energy Occurs in Different Forms

Guided Notes · Reader p. 3

Write or draw one example that shows a change for each form.

FORM	ONE EXAMPLE THAT SHOWS A CHANGE
motion	
sound	
light	
heat	
electrical energy	

Vocabulary — Stored Energy

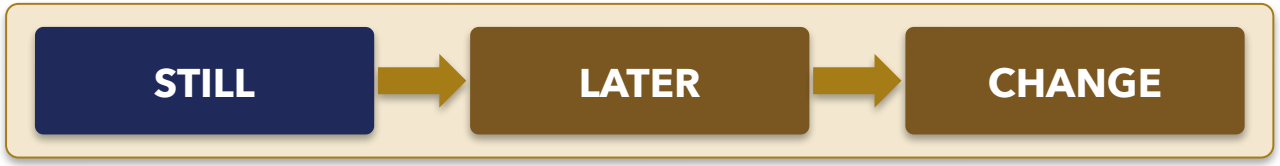
Reader p. 4

VOCABULARY

TEACH

stored energy

energy that has the ability to cause change at a later time



Reader p. 4

WRITE N

V2

Vocabulary – Stored Energy

Guided Notes · Reader p. 4

stored energy – energy that has the ability to cause change at a _____ time.

Write the missing word in the blank.

CHECK N

V2

Vocabulary – Stored Energy

Guided Notes · Reader p. 4

stored energy – energy that has the ability to cause change at a **later** time.

later

PREDICT R

Does a still battery have energy?

Reader p. 4 — a battery sits still and shows no change… yet.

A battery sitting still shows no change — yet.

Inside a battery, chemicals store energy.
When it goes in a flashlight, the stored energy is released.

R

The stored energy of position

Reader p. 5 — an object held up can fall and cause a change.

Raised apple → let go → it falls

An apple held above the ground has stored energy because of its position. Let it go and it will fall.

Stored energy can cause a change later



Stretch a rubber band — it now stores energy.
Let it go — it moves and makes a sound: the change.

N3

Objects Can Store Energy

Guided Notes · Reader pp. 4–5

STORED ENERGY → RELEASE → CHANGE

EXAMPLE	WHAT CHANGE COULD HAPPEN LATER?
battery (in a flashlight)	
stretched rubber band	
raised object (apple)	

Objects also have stored energy because of their _____ .

N3 Objects Can Store Energy

Guided Notes · Reader pp. 4–5

STORED ENERGY → **RELEASE** → **CHANGE**

EXAMPLE	WHAT CHANGE COULD HAPPEN LATER?
battery (in a flashlight)	
stretched rubber band	
raised object (apple)	

Objects also have stored energy because of their **position** .

position

EVIDENCE R

Recognizing evidence

Reader pp. 1–6 — we see the change, not the energy.

Energy can be invisible.
We recognize energy by the change we observe.

Evidence = the change you see, hear, or feel.

N4

Recognizing Evidence

Guided Notes · Reader pp. 1–6

Draw a change you observed. Show what it looked like BEFORE and AFTER.

BEFORE	AFTER
draw	draw

What changed? _____

N4

Check Your Thinking

Guided Notes · Reader pp. 1–6

CHECK YOUR THINKING

How do we know energy is involved if we cannot see energy itself?

Start with: "We know because we see …"

CHECK C

We know because we see a change

SAMPLE

"We know because we see a change — like a ball starting to move or a light turning on."

Any change you observe is evidence of energy.

EVIDENCE C

Which change shows energy at work?

A toy car starts to roll.

A rock keeps sitting still.

A room stays dark.

Look for a change — that is the evidence of energy.

WRITE N

BQ

BQ · Now I Think

Guided Notes p. 3

BIG QUESTION

Where can we observe evidence of energy causing change?

Now I think...

C

Review 1 – concept

Energy is best described as –

A the ability to cause change

B a visible glow

C something only batteries have

D a kind of fuel

CHECK C

Review 1 – answer

Energy is best described as –

A **the ability to cause change**

B a visible glow

C something only batteries have

D a kind of fuel

Energy is the ability to cause change (Reader p. 1).

C

Review 2 — visual

A pencil is dull, then you sharpen it. What change do you observe?

A It gets longer

B It gets shorter and sharper

C It changes color

D It disappears

CHECK C

Review 2 — answer

A pencil is dull, then you sharpen it. What change do you observe?

A It gets longer

B It gets shorter and sharper

C It changes color

D It disappears

The pencil becomes shorter and sharper — a change.

C

Review 3 — scenario

Which object has stored energy that can cause a change LATER?

A A ball already rolling

B A toy already moving

C A stretched rubber band

D A flashlight already on

CHECK C

Review 3 — answer

Which object has stored energy that can cause a change LATER?

A A ball already rolling

B A toy already moving

C **A stretched rubber band**

D A flashlight already on

The rubber band stores energy; it changes when released.

C

Review 4 — comparison

A carousel spins and plays music. Which two forms of energy do you observe?

A motion and silence

B sound and stillness

C only light

D motion and sound

CHECK C

Review 4 — answer

A carousel spins and plays music. Which two forms of energy do you observe?

A motion and silence

B sound and stillness

C only light

D **motion and sound**

The spinning is motion; the music is sound (Reader p. 3).

C

Review 5 — concept

How do we know energy is involved if we cannot see it?

A Energy can always be seen

B We observe a change happen

C We guess without evidence

D Nothing can change

CHECK C

Review 5 — answer

How do we know energy is involved if we cannot see it?

A Energy can always be seen

B We observe a change happen

C We guess without evidence

D Nothing can change

We see a change — like movement — as evidence.

C

Review 6 — scenario

A battery sits still in a flashlight. Does it have energy?

A No — still objects have no energy

B Only when it is moving

C Yes — stored energy that can cause change later

D Only when the flashlight is on

CHECK C

Review 6 — answer

A battery sits still in a flashlight. Does it have energy?

A No — still objects have no energy

B Only when it is moving

C Yes — stored energy that can cause change later

D Only when the flashlight is on

A battery stores energy; it changes when released.